Kamil, Ridwan Kehinde

Scientist | Strategic Data Intelligence Analyst | Data Engineer

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United Kingdom

PROFESSIONAL SUMMARY

Results-driven **Scientist** and **Analytics Professional** with experience in big data analytics for predictive modelling, designing reliable data engineering pipelines, integrating complex operational datasets, and architecting governed analytical solutions for decision-making. Skilled in transforming large, multi-source raw datasets into trusted, reusable data products through automated ingestion (ETL/ELT), schema validation, robust data modelling, forecasting analytics using machine learning and strict quality assurance.

CORE TECHNICAL SKILLS

- **Programming and Data Analysis:** Python, Pandas, NumPy, PySpark, scikit-learn, TensorFlow/Keras, SQL, T-SQL, PostgreSQL, MySQL, SQLite, Advanced Excel and Power BI.
- **Predictive Modelling:** Time-series forecasting, regression, classification, ensemble learning, deep learning, feature engineering, model benchmarking and rolling-origin validation.
- **Data Engineering:** REST APIs, ETL/ELT pipelines, FastAPI, SQLAlchemy, batch processing, dimensional modelling, star schemas, data warehousing, validation, lineage, idempotent processing and automated ingestion.
- **Data Visualisation & Insight Delivery:** Power BI, Plotly, Chart.js, Leaflet, dashboard development, KPI tracking, geospatial analysis and stakeholder-focused data storytelling.
- **Cloud Deployment:** Azure App Service, Azure SQL Database, Microsoft Fabric, Databricks, OneLake, Delta Lake, Unity Catalog, Supabase, GitHub Actions, Docker and CI/CD.
- **Whole Energy System Analytics:** Electricity and gas demand, transport, flood and river monitoring, generation availability, grid congestion, constraint costs, wind forecasting, operating margin, carbon intensity, flexibility, green hydrogen and Net Zero analysis.
- **Delivery and Governance:** Requirements gathering, stakeholder engagement, data governance, quality assurance, audit logging, pipeline monitoring, technical documentation, Agile delivery and evidence-based recommendations.

PROFESSIONAL EXPERIENCE

Scientist (Engineering & Analytics) — 09/2023 – Present | Lancashire, United Kingdom

Centre for Data and Complex Systems Research, Edge Hill University

Role Description – Based in the Department of Computer Science and Engineering, I design and build robust data pipelines and analytics systems. I leverage data engineering and data analytics expertise to drive modelling, forecasting, research and strategic decision-making for infrastructure planning and stakeholder engagement.

Key Contributions –

- Developed deep learning forecasting models for a hybrid power system with 95% accuracy, improving the reliability of generation forecasts and supporting long-term energy-use planning through structured data preparation, feature engineering and reproducible modelling pipelines.
- Analysed 16 years of Great Britain electricity generation and system data, using Python, SQL and Power BI to process and integrate large datasets, identify trends in demand variability, renewable penetration, carbon intensity and system stress, and translate findings into stakeholder-focused web dashboards.
- Delivered interactive Streamlit/Render web-based forecasting dashboards, connecting processed analytical datasets and model outputs to support facility management in forecasting future hybrid power-system generation to support operational decisions.
- Designed interactive Power BI, Plotly Dash and Streamlit dashboards to transform 5 years of half-hourly energy-generation data into governed KPIs, trends, risk indicators and decision-ready evidence for technical and non-technical audiences.
- Gathered, structured and managed large datasets relating to energy generation output, carbon intensity and system stress, improving data accuracy and analytical efficiency by 30% through SQL-based data processing, validation and structured database workflows.
- Integrated NASA meteorological datasets using QGIS and structured data-integration workflows, increasing the accuracy of long-term energy forecasts by 20%.
- Introduced reproducible data and model-development pipelines and quality-assurance practices, including chronological validation, baseline benchmarking, automated data checks, Git version control, documented feature pipelines and cloud deployment workflows.
- Maintained 100% GDPR-compliant data handling across local management, public and private datasets, ensuring responsible data use.

--- Data Engineering and Analytics Portfolio Projects---

The Projects listed in this section are available at my **Dedicated Website and Projects Repository Link:**

- **Greater Manchester Transport Intelligence Platform:** Developed and deployed a production-grade public transport data engineering platform using *Python, Azure SQL Database, GitHub Actions, Microsoft Fabric Data Factory, Fabric Lakehouse, OneLake, FastAPI, SQLite and Azure App Service*, automating TfGM GTFS ingestion, source-version preservation, validation, dimensional modelling, governed analytics, automated 15 mins publication-change detection and cloud delivery of route, stop, operator and network intelligence across over 3 millions of timetable records records. ***(Output: Deployed Web Platform)***
- **Greater Manchester Flood & River Monitoring ETL and Reporting Platform:** Developed and deployed an end-to-end environmental data engineering and operational intelligence platform using *Python, Azure SQL Database, SQL Server, SSIS, T-SQL, SSRS, GitHub Actions, Microsoft Azure and Power BI*, automating near-real-time ingestion of Environment Agency river-level and flood-warning data through validated ETL pipelines, dimensional modelling and governed cloud storage. Delivered an interactive Power BI dashboard providing operational intelligence on monitored stations, river-level trends, high-risk locations, active flood warnings and ETL performance to support evidence-led environmental monitoring, resilience planning and public-sector decision making across Greater Manchester. ***(Output: Live Monitoring BI Platform)***

- **Great Britain Electricity Generation Availability Live Analytics Dashboard:** Developed and deployed an end-to-end generation availability intelligence platform using *Python, PySpark, Databricks, Unity Catalog, Delta Lake, PostgreSQL, FastAPI, GitHub Actions, Power BI and Microsoft Azure*, automating hourly ingestion of NESO/Elexon generation availability publications, preserving canonical publication history, analysing fuel, system and unit-level revisions, and delivering governed revision intelligence through a live web application and Power BI-ready analytical model. ***(Output: Live Analytics Dashboard)***
- **Great Britain Grid Congestion and Constraint Cost Intelligence Hub:** Developed and deployed an Azure-based energy analytics and machine-learning platform using *Azure SQL Database, Flask, GitHub Actions and Azure App Service*, integrating NESO congestion, thermal constraint cost and ETYS datasets to analyse grid stress, cost patterns and high-cost event risk, achieving a ROC AUC of 0.878. ***(Output: Live-Hub link)***
- **Great Britain Operating Margin Intelligence Dashboard:** Designed and deployed an end-to-end cloud analytics application using *Python, PostgreSQL, Supabase, Plotly Dash, GitHub Actions and Microsoft Azure*, transforming open electricity-system data into automated insights on operating margin, reserve sufficiency, imbalances and system readiness. ***(Output: Live Intelligence Dashboard)***
- **GB Wind Energy Generation Forecast Monitoring Dashboard:** Developed and deployed a cloud based NESO Wind Forecast Revision Monitor using *Python, Supabase PostgreSQL, GitHub Actions and Streamlit*, automating the collection of Great Britain's 14 day ahead wind forecast data and enabling live visualisation of forecast trends and revision behaviour. ***(Output: Live Streaming Dashboard)***
- **Hybrid Power Generation Forecasting App:** Developed and deployed a *BiLSTM-Informer deep learning model* using six years of hourly PV generation and meteorological data to deliver multi-step forecasts for Edge Hill University's 39.02 kWp hybrid PV-grid system, achieving *89%–97.3%* forecasting performance through a real-time web application. ***(Output: Deployed Forecasting Dashboard)***

Principal Energy Systems Analyst 06/2018 – 07/2023 | Abuja, Nigeria
National Agency for Science and Engineering Infrastructure (Federal Ministry of Science, Technology & Innovation Department)

Role Description - I contributed to and led analytical research and infrastructure projects focused on energy system performance monitoring, efficiency optimisation and sustainable energy deployment.

Key Achievements -

- Spearheaded an analytics-led project assessing public-sector adoption of renewable energy technologies, including photovoltaic and low-carbon systems, supporting interventions that contributed to a 25% increase in clean energy usage.
- Integrated IoT sensor data with monitoring dashboards to capture and analyse infrastructure telemetry, identifying consumption patterns, operational inefficiencies and opportunities for cost-saving retrofit interventions.
- Developed and evaluated energy-harvesting prototypes using an iterative, data-driven testing approach, analysing performance, reliability and feasibility data to refine engineering design assumptions.
- Analysed historical datasets on Virtual Power Plant deployment, identifying trends and patterns in decentralised energy systems to support infrastructure planning, power-system decentralisation and climate-action initiatives.
- Translated analytical findings into technical reports, performance insights and stakeholder presentations, improving understanding of complex energy-system data and supporting evidence-based engineering and policy decisions.

Design and Analytics Engineer 12/2016 – 05/2018 | Abuja, Nigeria
Julius Berger Constructions Plc.(Julius Berger International Subscidiary)

Role Description - Analytical design of building infrastructure projects, combining electrical modelling with data-driven project management to ensure operational integrity and regulatory compliance.

Key Achievements -

- Led the implementation of a data-driven risk assessment framework, using incident telemetry and BI dashboards to improve safety monitoring and strengthen adherence to the hierarchy of control.
- Translated stakeholder requirements into structured analytical outputs, improving the reliability of system design assumptions and supporting evidence-based project planning.
- Performed stochastic load-flow simulations and demand profiling using Advanced Excel to model infrastructure project scenarios, optimising M&E system design and ensuring full compliance with British Standards and Eurocodes.
- Coordinated testing, commissioning and technical documentation across multidisciplinary project teams, ensuring delivery against regulatory, quality and programme requirements.

EDUCATION

Ph.D. in Engineering, 09/2026 | Lancashire, United Kingdom
Edge Hill University
Research Focus: Energy Resources Forecasting using Deep-Learning Algorithms to optimise the Energy Management Systems.

M.Sc. in Electrical/Electronic Engineering, **Grade: Distinction** 10/2016 | London, United Kingdom
University of East London

B.Eng. in Electrical Engineering, **Grade: Second Class Upper** 12/2014 | Niger State, Nigeria
Federal University of Technology, Minna

PROFESSIONAL RECOGNITION & AWARDS

- **Certificate, 1st Edition of the Innovation Hub Challenge, 2024** World Energy Council in the Netherlands, Future Energy Leaders.
- **McKinsey Leadership, Business & Digital Skills, 2023.** Certificate of Completion.
- **Recipient, Two (2) Ph.D. Degree Scholarships, 2023.** Edgehill University and PTDF awarded me a full study scholarship.
- **Project Management, 2016.** Certificate of Proficiency. The UK Chartered Institute of I.T.
- **Certificate, 1st Edition of the Innovation Hub Challenge, 2024.** World Energy Council in the Netherlands, Future Energy Leaders.